

SYDNEY BOYS' HIGH SCHOOL

MOORE PARK, SURRY HILLS



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 1998

MATHEMATICS

3 UNIT ADDITIONAL (3/4 UNIT COMMON)

Time allowed: 3 Hours
(plus five minutes reading time)

Examiners: P.R. Bigelow & P.S. Parker

DIRECTIONS TO CANDIDATES

- ALL questions may be attempted.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Start each question in a new answer booklet. Indicate your name, class and teacher on each new booklet
- Additional answer booklets may be obtained from the supervisor upon request.

NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate Examination Paper for this subject.

Question 1 (Start a new page)

Marks

- (a) Find the value of a such that $P(x) = x^3 - 2x^2 - ax + 6$ is divisible by $x + 2$ 2
- (b) For a given series $T_{n+1} - T_n = 7$, $T_1 = 3$, find the value of S_{100} , where $S_n = T_1 + T_2 + \dots + T_n$. 2
- (c) The interval joining $A(x_1, y_1)$ and $B(x_2, y_2)$ is trisected by the points $P(-2, 3)$ and $Q(1, 0)$.
Write down the coordinates of A and B . 3
- (d) Find the acute angle (to the nearest degree) between the lines $x - y = 2$ and $2x + y = 1$. 2
- (e) Solve $|2x - 1| - |x| \leq 0$ 3

Question 2 (Start a new page)

Marks

(a) Find:

(i) $\int \frac{dx}{4+x^2}$

1

(ii) $\int_0^{\frac{\pi}{2}} \cos^2 \frac{t}{2} dt$

3

(b) Given $f(x) = \sin^{-1} 2x$

3

(i) Write down the domain and range of $y = f(x)$

(ii) Sketch the curve.

(iii) Find the exact value of $f'(0.25)$

(c) Solve $1 + \cos 2x = \sqrt{3} \sin 2x$ where $-\pi < x < \frac{\pi}{4}$

3

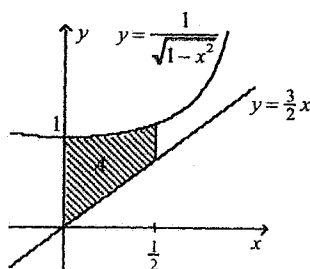
(d) Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x e^{\tan x} dx$ using the substitution $u = \tan x$

2

Question 3 (Start a new page)

Marks

- (a) (i) Show algebraically that the line $y = \frac{3}{2}x$ does not meet the curve with equation $y = \frac{1}{\sqrt{1-x^2}}$ 3
- (ii) Find the area of the region A , bounded by the curve $y = \frac{1}{\sqrt{1-x^2}}$ and the lines $x = 0$, $x = \frac{1}{2}$ and $y = \frac{3}{2}x$. 3

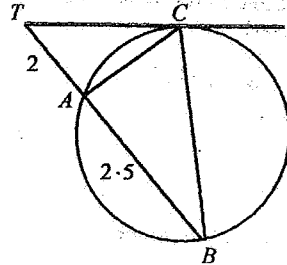


- (b) A spherical balloon is expanding so that its volume $V \text{ mm}^3$ increases at a constant rate of 72 mm^3 per second. What is the rate of increase of its surface area $A \text{ mm}^2$, when the radius is 12 mm. 2
- (c) Given $(3x-2)^{100} = a_{100}x^{100} + a_{99}x^{99} + \dots + a_1x + a_0$, where a_i ($i = 0, \dots, 100$) is a real number. Evaluate $a_{100} + a_{99} + \dots + a_1 + a_0$ 2
- (d) Differentiate $x^2 \cos^{-1} x$ 2

Question 4 (Start a new page)

Marks

(a)



TC is a tangent. $TA = 2$ units, $AB = 2.5$ units.

Find the length of TC .

2

(b) Find the coefficient of y^{10} in the expansion $(1 + y)(3y^2 - 2)^7$

3

(c) (i) Show that the equation of the tangent at $T(-2t, t^2)$ on the parabola $y = \frac{1}{4}x^2$ is given by $y + tx - t^2 = 0$

2

(ii) If the point $M(x, y)$ is the midpoint of the interval TA where A is the x intercept of the tangent at T . Find the equation of the locus of M as T moves on the parabola.

3

(d) Evaluate $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin x}$

2

Question 5 (Start a new page)

Marks

- (a) A particle is moving in a straight line with Simple Harmonic Motion. If the amplitude of the motion is 8 cm and the period of the motion is 6 seconds. 4
- (i) Express the displacement, x , of the particle as a function of time, t .
 - (ii) Calculate the maximum velocity of the particle.
 - (iii) Calculate the maximum acceleration of the particle.
 - (iv) Calculate the speed when it is 4 cm from the centre of the motion.
- (b) Twelve students sit around a circular table. 4
- (i) How many ways can they be arranged?
 - (ii) If 4 students wish to sit together, how many seating arrangements can be made?
 - (iii) Let three of the students be A , B and C . Find the probability that A does not sit next to either B or C .
- (c) A particle is projected from a point O . After 5 seconds its horizontal and vertical displacements from O are 60 m and 57.5 m respectively. 4
If the particle is still rising, find its initial velocity.
(You may take $g = 10 \text{ m/s}^2$)

Question 6 (Start a new page)

Marks

- (a) How many times should a die be thrown so that the probability of obtaining at least one multiple of 3 exceeds 0.95? 2
- (b) At any time t , the rate of cooling of the temperature T of a body when the surrounding temperature is S is given by the equation 5

$$\frac{dT}{dt} = -k(T - S), \text{ for some constant } k$$
 - (i) Show that $T = S + Ae^{-kt}$, for some constant A , satisfies this equation.
 - (ii) A metal rod has a temperature of 1390°C and cools to 1060°C in 10 minutes when the surrounding temperature is 30°C .
Find how much longer it will take the rod to cool to 110°C , giving your answer correct to the nearest minute.
- (c) A particle is moving along the x axis with velocity $v \text{ m s}^{-1}$, and acceleration $\ddot{x} \text{ m s}^{-2}$. 5
 - (i) Show that $\ddot{x} = \frac{d}{dx}(\frac{1}{2}v^2)$
 - (ii) If $v^2 = 24 - 6x - 3x^2$ find the acceleration of the particle at the particle's greatest displacement from the origin O .

Question 7

- (a) (i) Show that $P(x) = 8x^3 - 12x^2 + 6x + 13$ has only one zero x_1 , and that this zero is negative. 4
- (ii) Find the least value of c , where c is a positive integer, such that $-c < x_1 < 0$
- (iii) With $-\frac{c}{2}$ as a first approximation, find a better approximation to x_1 , using Newton's Method once. Express your answer correct to two decimal places.

- (b) Consider $\tan^{-1} y = 2 \tan^{-1} x$ 4
- (i) Express y as a function of x .
- (ii) Show that the function has no turning points.
- (iii) State the domain of the function.
- (iv) Sketch the graph of the function.

- (c) If $\tan \alpha$ and $\tan \beta$ are the two values of $\tan \theta$ which satisfy the quadratic equation: 4
- $$a \tan^2 \theta + b \tan \theta + c = 0$$
- (i) Find $\tan(\alpha + \beta)$
- (ii) Show that $\tan^2(\alpha - \beta) = \frac{b^2 - 4ac}{(a + c)^2}$

END OF THE PAPER



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Mathematics Extension 1

Sample Solutions

① a) let $x+2=0$

$x = -2$

So $P(-2) = -16 + 6 + 2a = 0$

$a = 5$ ②

b) $T_{n+1} - T_n = 7$

$T_2 - T_1 = 7$ $T_1 = 3$

$T_2 - 3 = 7 \Rightarrow T_2 = 10$

$T_3 - T_2 = 7$

$T_3 - 10 = 7 \Rightarrow T_3 = 17$

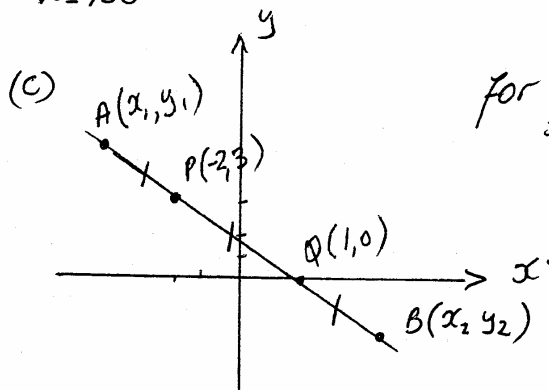
So $\{3+10+17+\dots\}$

$a=3$

$d=7$

$n=100$

$S_{100} = \frac{100}{2} \{6 + 99 \times 7\} = 34950$ ②



for AP, midpoint P.

$\frac{x_1+1}{2} = -2 \Rightarrow x_1 = -5$

$\frac{y_1+0}{2} = 3 \Rightarrow y_1 = 6$

$A(-5, 6)$

for PB, midpoint Q.

$\frac{-2+x_2}{2} = 1 \Rightarrow x_2 = 4$ ③

$\frac{3+y_2}{2} = 0 \Rightarrow y_2 = -3$

$B(4, -3)$

(d) $x-y=2$ $m_1=1$

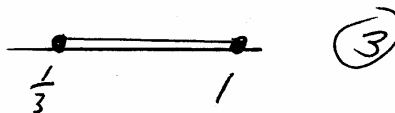
$2x+y=1$ $m_2=-2$

$\tan \theta = \left| \frac{1+2}{1-2} \right|$ $\theta \doteq 72^\circ$ ②

(e) $|2x-1| \leq |x|$

$2x-1 \leq x$ or $2x-1 \geq -x$

$x \leq 1$ or $x \geq \frac{1}{3}$



$$1) (i) \int \frac{dx}{4+x^2} = \frac{1}{2} \tan^{-1} \frac{x}{2} + C$$

$$3) (ii) \int_0^{\frac{\pi}{2}} \cos^2 \frac{t}{2} dt = \int_0^{\frac{\pi}{2}} \frac{1}{2} (\cos t + 1) dt$$

$$= \frac{1}{2} \left[\sin t + t \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\sin \frac{\pi}{2} + \frac{\pi}{2} \right) - (0 + 0) \right]$$

$$= \frac{\pi}{4} + \frac{1}{2}$$

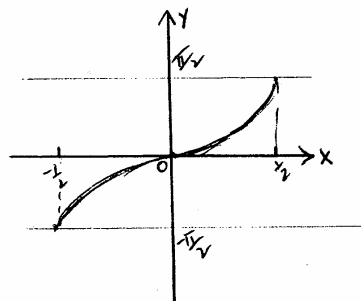
$\downarrow$
 $\frac{1}{2}(\cos t + 1)$

$$b) (i) f(x) = \sin^{-1} 2x$$

domain: $-1 \leq 2x \leq 1$
ie $-\frac{1}{2} \leq x \leq \frac{1}{2}$

range: $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$

1(ii)



$$1) (iii) f'(x) = \frac{1}{\sqrt{1-4x^2}} \cdot 2x$$

$$f'\left(\frac{1}{4}\right) = \frac{2}{\frac{2}{\sqrt{3}}} = \frac{4}{\sqrt{3}}$$

or $\frac{4\sqrt{3}}{3}$

$$-\pi < x < \frac{\pi}{4}$$

(c)

$$3) 1 + \cos 2x = \sqrt{3} \sin 2x$$

$$\cos 2x - \sqrt{3} \sin 2x = -1 \quad \text{--- (A)}$$

$$R \cos(2x + \alpha) = -1$$

$$\cos \alpha = 1 \text{ and } \sin \alpha = \sqrt{3}$$

$$\tan \alpha = \sqrt{3} \Rightarrow \boxed{\alpha = \frac{\pi}{3}}$$

$$R = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$\therefore$ (A) becomes

$$2 \cos\left(2x + \frac{\pi}{3}\right) = -1 \quad \text{for } -\frac{5\pi}{3} < 2x + \frac{\pi}{3} < \frac{5\pi}{6}$$

$$\cos\left(2x + \frac{\pi}{3}\right) = -\frac{1}{2}$$

$$2x + \frac{\pi}{3} = \frac{2\pi}{3}, -\frac{2\pi}{3}, -\frac{4\pi}{3}$$

$$\Rightarrow x = \frac{\pi}{6}, -\frac{\pi}{2}, -\frac{5\pi}{6}$$

(d)

$$u = \tan x, \quad du = \sec^2 x dx$$

2

$$\text{When } x=0, u=0$$

$$\text{When } x=\frac{\pi}{4}, u=\tan \frac{\pi}{4} = 1.$$

$$\therefore I = \int_0^1 e^u du$$

$$= [e^u]_0^1$$

$$= e^1 - e^0$$

$$= e - 1$$

(a)(i) Curves have a simultaneous solution if they meet.

$$\therefore y = \frac{1}{\sqrt{1-\frac{4y^2}{9}}} \quad \text{or} \quad \frac{9x^2}{4} = \frac{1}{1-x^2}$$

$$9y^2 - 4y^4 - 9 = 0$$

$$\Delta = 81 - 144$$

$$< 0$$

$$9x^2 - 9x^4 - 4 = 0$$

$$\Delta = 81 - 144$$

$$< 0$$

$\therefore$ No real solutions for $y^2(ax^2)$, so no real solutions for $y(ax)$

$\therefore$ No point of intersection.

$$\begin{aligned} \text{(ii) Area} &= \int_0^{1/2} \left\{ \frac{1}{\sqrt{1-x^2}} - \frac{3x}{2} \right\} dx \\ &= \left[\sin^{-1} x - \frac{3x^2}{2} \right]_0^{1/2} \\ &= \frac{\pi}{6} - \frac{3}{16} \end{aligned}$$

$$\text{(b) } \frac{dV}{dt} = 72 \text{ mm}^3/\text{s}$$

$$V = \frac{4}{3}\pi r^3$$

$$A = 4\pi r^2$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dA}{dr} = 8\pi r$$

$$\begin{aligned} \frac{dA}{dt} &= \frac{dA}{dr} \cdot \frac{dr}{dV} \cdot \frac{dV}{dt} \\ &= \frac{8\pi r}{4\pi r^2} \cdot 72 \end{aligned}$$

$$\text{When } r=12, \quad \frac{dA}{dt} = 12 \text{ mm}^2/\text{s}$$

$$\text{(c) } (3x-2)^{100} = a_{100}x^{100} + a_{99}x^{99} + a_{98}x^{98} + \dots + a_1x + a_0$$

$$\text{When } x=1$$

$$\begin{aligned} \text{L.H.S.} &= (3-2)^{100} \\ &= 1 \end{aligned}$$

$$\text{R.H.S.} = a_{100} + a_{99} + a_{98} + \dots + a_1 + a_0$$

$$\therefore a_{100} + a_{99} + \dots + a_0 = 1$$

$$\text{(d) } D_x(x^2 \cos^{-1} x) = 2x \cos^{-1} x - \frac{x^2}{\sqrt{1-x^2}}$$

Q (4)

(a)

$$T_C^2 = T_B \cdot T_A$$

$$= 4.5 \times 2$$

$$= 9$$

$$\therefore T_C = 3 \quad \text{--- (1)}$$

(b) To find coeff. of y^{10} , we need:

(i) coefficient of y^9 times 1

(ii) coefficient of y^{10} in $(3y^2-2)^7$ times 1.

Now, $(3y^2-2)^7$

$$T_{r+1} = \binom{7}{r} (3y^2)^{7-r} (-2)^r$$

$$= \binom{7}{r} 3^{7-r} (-2)^r \cdot y^{14-2r}$$

(a) $14-2r = 9 \Rightarrow r = \frac{5}{2}$ (not possible)

(b) $\therefore 14-2r = 10 \Rightarrow 2r = 4$ i.e. $r = 2$.

$\therefore$ coefficient of y^{10} .

$$\binom{7}{2} 3^5 (-2)^2 = 20412 \quad \text{--- (3)}$$

(c). (i) $y = x^2/4$, $\frac{dy}{dx} = x/2$, $\frac{dy}{dx} \Big|_{x=-2t} = -t$

$\therefore$ equation of tangent, $y - t^2 = -t(x + 2t)$.

$\Rightarrow tx + y + t^2 = 0 \quad \text{--- (2)}$

When $y = 0$ $x = -t \Rightarrow$ A $(-t, 0)$.
T $(-2t, t^2)$

M $= \left(-\frac{3t}{2}, \frac{t^2}{2}\right)$. $|$ $x = -\frac{3t}{2} \therefore t = -\frac{2x}{3}$.

$y = \frac{1}{2} \left(\frac{4x^2}{9}\right) \Rightarrow$ locus $x^2 = \frac{9y}{2} \quad \text{--- (3)}$

(d).

$$\lim_{x \rightarrow 0} \frac{5x(1-2\sin^2 x)}{\sin x} = 5 \lim_{x \rightarrow 0} \frac{1}{\left(\frac{\sin x}{x}\right)} = 5 \lim_{x \rightarrow 0} x \sin x = 5 \quad \text{--- (2)}$$

tion 5

(i) $x = a \cos(nt + \alpha)$
 $a = 8$
 $\frac{2\pi}{n} = 6$
 $n = \frac{\pi}{3}$
 $x = 8 \cos(\frac{\pi}{3}t + \alpha) \text{ cm}$

(ii) $v^2 = n^2(a^2 - x^2)$
 Max velocity when $x = 0$
 $\therefore v^2 = (\frac{\pi}{3})^2 \cdot 64$
 $\therefore v = \pm \frac{8\pi}{3} \text{ cm/s}$

(iii) $\ddot{x} = -n^2x$
 Max acceleration when $x = \pm 8$
 $\ddot{x} = -(\frac{\pi}{3})^2 \cdot \pm 8$
 $= \pm \frac{8\pi^2}{9} \text{ cm/s}^2$

(iv) $v^2 = n^2(a^2 - x^2)$
 $v^2 = (\frac{\pi}{3})^2 (8^2 - 4^2)$
 $v^2 = \frac{\pi^2}{9} \cdot 48$
 $\therefore v = \pm \frac{4\pi\sqrt{3}}{3} \text{ cm/s}$

(b) (i) $11!$ 1
 (ii) $8! \times 4!$ 1
 (iii) $72 \times 9!$ 2

Probability is $\frac{72 \times 9!}{11!}$
 $= \frac{72}{110}$
 $= \frac{36}{55}$

(c) $\ddot{x} = 0$ $\ddot{y} = -10$
 $\dot{x} = v \cos \theta$ $\dot{y} = -10t + v \sin \theta$
 $x = v \cos \theta t$ $y = -5t^2 + v \sin \theta t$

$v \cos \theta = \frac{x}{t}$ $v \sin \theta = \frac{y + 5t^2}{t}$
 When $t = 5$: $v \cos \theta = 12$ $v \sin \theta = \frac{57.5 + 125}{5}$
 $= 36.5$

$\therefore v^2 = 12^2 + 36.5^2$
 $= 1476.25$

$\therefore v = 38.42199 \dots$
 $\approx 38.4 \text{ m/s}$

(Unfortunately this is on the down ward flight not upward)

39916800
 967680
 26127360

$$6 \text{ (a)} \quad P(\text{multiple of 3 in a single throw}) = \frac{2}{6} = \frac{1}{3}$$

$$P(\text{not a multiple of 3}) = \frac{2}{3}$$

$$P(\text{no multiples of 3 in } n \text{ tosses}) = \left(\frac{2}{3}\right)^n$$

$$\text{We need } \left(\frac{2}{3}\right)^n < 0.05$$

$$n > \frac{\log(0.05)}{\log\left(\frac{2}{3}\right)} \quad 2$$

$$n > 7.38$$

$\therefore$ 8 tosses are needed.

(i)

$$\frac{dT}{dt} = -K(T-S)$$

$$T = S + Ae^{-Kt}$$

$$\frac{dT}{dt} = -KAe^{-Kt}$$

$$\frac{dT}{dt} = -K(T-S)$$

(ii)

$$T = S + Ae^{-Kt}$$

$$S = 30$$

When $t = 0$, $T = 1390$

$$1390 = 30 + A$$

$$A = 1360$$

$$T = 30 + 1360 e^{-Kt}$$

①

When $t = 10$

$$T = 1060$$

$$1060 = 30 + 1360 e^{-10K}$$

$$\frac{1030}{1360} = e^{-10K}$$

$$K = -\frac{1}{10} \ln\left(\frac{1030}{1360}\right)$$

②

$$K = 0.02779 \dots$$

When $T = 110$ $t = ?$

$$110 = 30 + 1360 e^{-Kt}$$

$$\frac{80}{1360} = e^{-Kt}$$

$$t = -\frac{1}{K} \ln\left(\frac{80}{1360}\right)$$

$$t = 101.9$$

4

92 mins later.

(c)(i) RTP $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$\begin{aligned}
 \text{RHS} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 &= \cancel{v} \times \frac{dv}{dx} \quad \text{since } v \text{ is a function of } x. \\
 &= \frac{dx}{dt} \times \frac{dv}{dx} \\
 &= \frac{dv}{dt} = \ddot{x} = \text{LHS}
 \end{aligned}$$

2

(ii) $v^2 = 24 - 6x - 3x^2$

$$\frac{1}{2} v^2 = 12 - 3x - \frac{3}{2} x^2$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -3 - 3x \quad |$$

Greatest displacement when $v = 0$

$$24 - 6x - 3x^2 = 0$$

$$x^2 + 2x - 8 = 0 \quad |$$

$$(x+4)(x-2) = 0$$

$$x = -4, 2$$

Acceleration at maximum displacement

$$\text{is } \ddot{x} = -3 - 3(-4) = 9 \text{ m/s}^2 \quad |$$

$$\text{or } \ddot{x} = -3 - 3(2) = -9 \text{ m/s}^2$$

7 (a) (i) $P(x) = 8x^3 - 12x^2 + 6x + 13$

$$P'(x) = 24x^2 - 24x + 6$$

$$= 6(4x^2 - 4x + 1)$$

$$= 6(2x-1)^2$$

Start pt where $P'(x) = 0$

$$(2x-1)^2 = 0$$

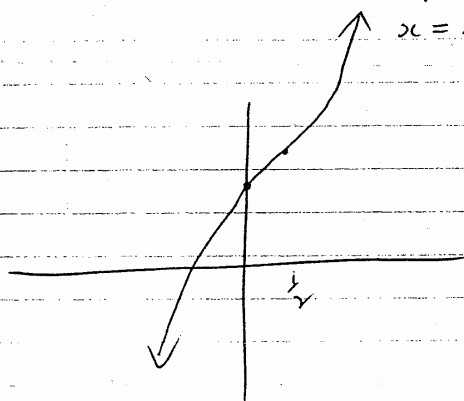
$$x = \frac{1}{2}$$

$$P(0.4) > 0 \text{ and } P(0.6) > 0$$

Horizontal pt of inflexion where

$$x = \frac{1}{2}, y = 1 - 3 + 3 + 13 = 14$$

$$\text{When } x = 0, y = 13$$



From sketch it is clear that there is only one root which is negative

(ii) $P(1) = -8 - 12 - 6 + 13 < 0$

$\therefore$ Req'd value of c is $c = 0$

(iii)
$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = -\frac{1}{2} - \frac{f(-\frac{1}{2})}{f'(-\frac{1}{2})}$$

$$= -0.75$$

(b) (i) $\tan^{-1} y = 2 \tan^{-1} x$

Take tan of both sides

$$y = \tan(2 \tan^{-1} x)$$

$$y = \tan 2\alpha \quad \text{where } \alpha = \tan^{-1} x$$

$$y = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$y = \frac{2x}{1 - x^2}$$

(ii) $\frac{dy}{dx} = \frac{(1-x^2)2 - 2x(-2x)}{(1-x^2)^2}$

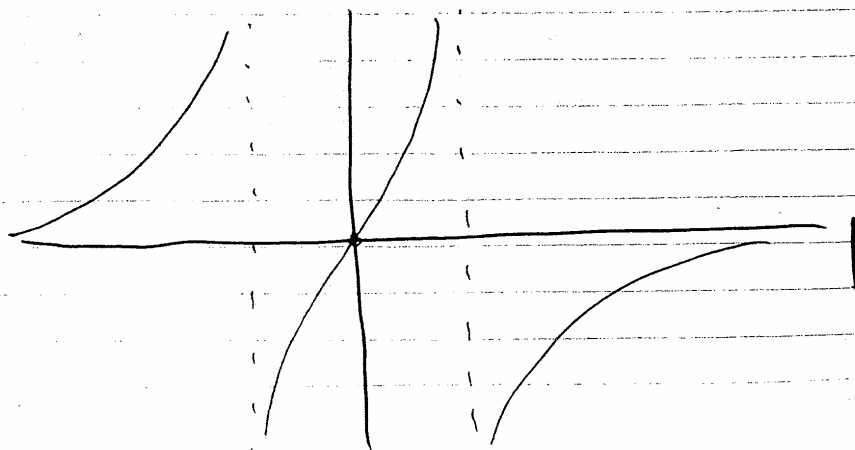
$$= \frac{2 - 2x^2 + 4x^2}{(1-x^2)^2}$$

$$= \frac{2x^2 + 2}{(1-x^2)^2} > 0$$

No turning pts

(iii) D: all real x except $x=1$ or $x=-1$

(iv)



$$(c) \quad a \tan^2 \theta + b \tan \theta + c = 0$$

$$(i) \quad \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \quad | \quad 2$$

$$= \frac{-\frac{b}{a}}{1 - \frac{c}{a}} = \frac{-b}{a - c} \quad |$$

$$(ii) \quad \tan^2(\alpha - \beta) = \left(\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \right)^2$$

$$= \frac{\tan^2 \alpha + \tan^2 \beta - 2 \tan \alpha \tan \beta}{(1 + \tan \alpha \tan \beta)^2} \quad |$$

$$= \frac{(\tan \alpha + \tan \beta)^2 - 4 \tan \alpha \tan \beta}{(1 + \tan \alpha \tan \beta)^2}$$

$$= \frac{\left(-\frac{b}{a}\right)^2 - 4 \frac{c}{a}}{\left(1 + \frac{c}{a}\right)^2}$$

$$= \frac{b^2 - 4ac}{(a + c)^2} \quad 2$$